ARCHITECTURE DECISION RECORD

ADR-012: Use SQS FIFO for order events

STATUS	SUPERSEDES
Accepted since 2026-11-07	**ADR-004** no longer holds

Deciders: Platform lead, Billing lead, Orders team

Context

Order events reach billing, email and analytics through a table polled every five seconds. The polling costs more than the reads it serves, and billing sees an order up to ten seconds late.

Decision drivers

- The events of one order arrive in the order they happened.
- Nothing new for the team to run at night.
- Under 500 $ a month at today's volume.

Considered options

1. Kafka
2. SQS FIFO (chosen)
3. Keep polling

Kafka

Good

- Replays any day of events
- Analytics reads it directly

Bad

- Brokers to size and watch
- Three times the cost

SQS FIFO (chosen)

Good

- Nothing to run
- Billing already reads SQS

Bad

- No replay once a message is read

Keep polling

Good

- No new system

Bad

- Five times the load on the database

Decision

> **Chosen: SQS FIFO**
> It meets every driver; replay, the one thing it lacks, no consumer needs today.

Consequences

Good, because

- Billing sees an order within a second
- The polled table can be dropped

Bad, because

- A consumer that fails mid-group holds that order's later events

Neither, but worth knowing

- Events older than four days are gone; analytics keeps its own copy

Links

- Decision paper
- ADR-004 (polling the orders table)